Adán Alejandro Sánchez

Backend & Infrastructure Engineer | Node.js · Kubernetes · Proxmox · PostgreSQL · MongoDB

San Luis Potosí, México · Linkedin · +52 444 819 66 81· adan.sanchez@proton.me · Web

Backend & Infrastructure Engineer experienced in designing, developing, and operating production software platforms. Former Backend Lead with a track record of designing scalable backend architectures, improving API response times by ~30%, and building business platforms that reduced operational costs by ~15%. Currently focused on modernizing legacy environments through Docker, Kubernetes, Talos Linux, and highly available Proxmox infrastructure with LINSTOR/DRBD storage replication. Experienced across the full software lifecycle, from software architecture and development to infrastructure design, deployment, and production operations.

PROFESSIONAL EXPERIENCE

Bitcat Services — **San Luis Potosí, México**
Software & infrastructure engineer — *May 2024–Present*

- Designed and deployed server infrastructure, including physical installation, cabling, network configuration and commissioning of a cluster of servers, switches, virtualization and storage.
- Managed highly available Proxmox clusters hosting production applications and databases, using LINSTOR-backed replicated storage to minimize downtime and VM migration times.
- Administered Kubernetes clusters running on Talos Linux, managing application deployments, highly available workloads, persistent storage, and production infrastructure.
- Implemented Cloudflare Tunnels to securely expose web services to the Internet without requiring static public IP addresses or direct inbound port exposure.
- Implemented CI/CD practices to improve software delivery, reducing build errors, time responses and deployments around 80% by using Github Actions & Docker.
- Built a full-stack project budgeting platform integrating internal services with the CROL API, reducing material-request lead time by ~40% and improving real-time budget visibility.
- Integrated web systems with PLCs inside network cabinets for maintenance, monitoring and Industry 4.0 environments.
- Dockerized existing systems, reducing around 50% errors related to non standardized development environments.
- Led the development of an internal quotation management platform, eliminating licensing costs associated with the previous proprietary solution.

Cooperative Computing — **San Luis Potosí, México, Remote**
Full Stack Intern — *September 2023–May 2024*

- Development of visual interfaces to improve user's experience of proprietary applications.
- Implementation of automation with GitHub, optimizing code's formatting and implementing custom repository's code standard rules.

AIA Partners — **San Luis Potosí, México**
Backend Lead Developer — June 2022–September 2023

- Implemented MongoDB replica sets to improve database availability, redundancy, and fault tolerance.
- Designed and implemented the backend architecture for a fintech platform using Node.js and Express, reducing API response times by ~30% and application errors by ~40%.

- Implemented scheduled jobs to automate backend workflows and recurring business processes.
- Implemented encryption and secure file-storage mechanisms for sensitive information.
- Provided technical support and training to users and administrators.
- Led and mentored the development team, coordinating responsibilities and technical training to accelerate product delivery.

Píxel & Byte **San Luis Potosí, México**
Social Service June 2022–December 2022

- Development of a DICOM image to JSON converter without loss of data. Enhancing AI's training to detect diseases in early stages .
- Development of a Node.js server capable of executing Python scripts, improving the speed of conversions and allowing to handle data up to 10 times bigger.

Universidad Politécnica de San Luis Potosí **San Luis Potosí, México**
Freelance videogame development November 2021–March 2022

- Development of an educational videogame with Javascript and RPG Maker, focused on teaching marketing processes for the "Semana LAG 2022".
- Implementation of Javascript scripts that allows the user's interface to work correctly, optimizing and improving the user's experience.

EDUCATION

UNIVERSIDAD AUTÓNOMA DE SAN LUIS POTOSI **San Luis Potosí, México**
Intelligent Systems Engineering. *June 2023*

TECHNICAL SKILLS

- Backend: Node.js, Express, NestJS, Laravel.
- Infrastructure & DevOps: Linux, Docker, Kubernetes, Talos, Proxmox VE, LINSTOR/DRBD, High Availability, Git, Nginx, GitHub Actions, CI/CD.
- Databases: MongoDB, PostgreSQL, MySQL, MariaDB.
- Frontend: React, Angular, JavaScript, TypeScript, HTML, CSS, TailwindCSS.

CERTIFICATIONS

- Docker & Kubernetes: The Practical Guide - Udemy (2026)
- Nest: Scalable Backend Development – Udemy (2025)
- React the complete guide – Udemy (2023)
- Agile hands on workshop – ANUIES TIC (2023)
- Blockchain Seminary – ANUIES TIC (2023)